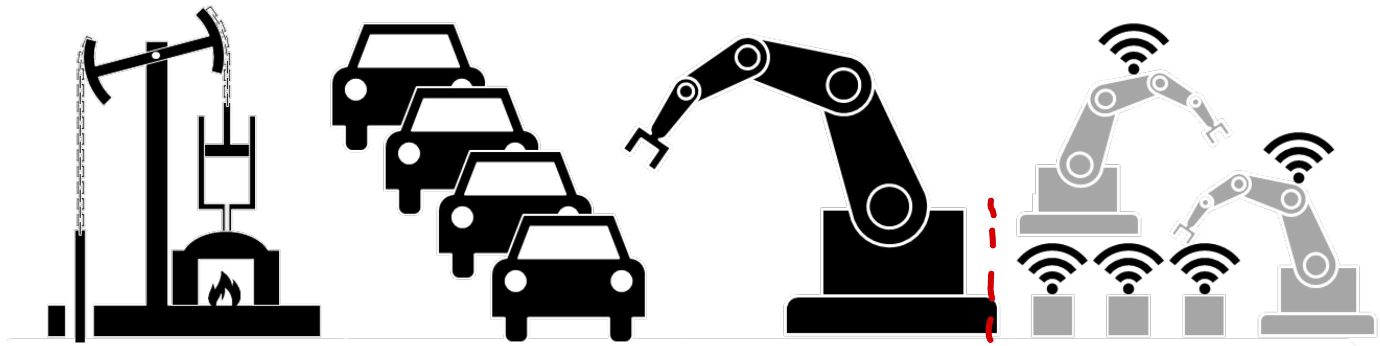
Smart Manufacturing

INDUSTRY 4.0

1760-1830	1870-1914	1950-1995	2014-
1st	2nd	3rd	4th
Mechanization, water power, steam power	Mass production, assembly line, electricity	Computer and automation	Cyber Physical Systems



(UNTIL NOW)

4th INDUSTRIAL REVOLUTION !

Research and industrial application

INDUSTRIAL IoT

Other names used in different countries:

- Smart Manufacturing
- Smart Factories
- Industrial Internet of Things

many evolution

ITALIAN INDUSTRY 4.0 ENABLING TECHNOLOGIES

Advanced Manufact. Solutions	•	Robot collaborativi interconnessi e rapidamente programmabili
Additive Manufacturing	•	(as STUDIED) → DIGITAL MANIF. create obj automatic Stampanti in 3D connesse a software di sviluppo digitali complex process
Augmented Reality	•	devices that enable to mix virtual and real vision, overlap Realtà aumentata a supporto dei processi produttivi real world with more INFO → helps human OPERATION / MAINTENANCE
Simulation	•	Simulazione tra macchine interconnesse per ottimizzare i processi
Horizontal/ Vertical Integration	•	huge set... Integrazione informazioni lungo la catena del valore dal fornitore al consumatore
Industrial Internet	•	Comunicazione multidirezionale tra processi produttivi e prodotti
Cloud	•	Gestione di elevate quantità di dati su sistemi aperti
Cyber-security	•	Sicurezza durante le operazioni in rete e su sistemi aperti
Big Data and Analytics	•	Analisi di un' ampia base dati per ottimizzare prodotti e processi produttivi HIGH DATA!

among all topics → BOX GEOM Represent. from X-RAY topography → 100GB repres.

[Source: MISE]

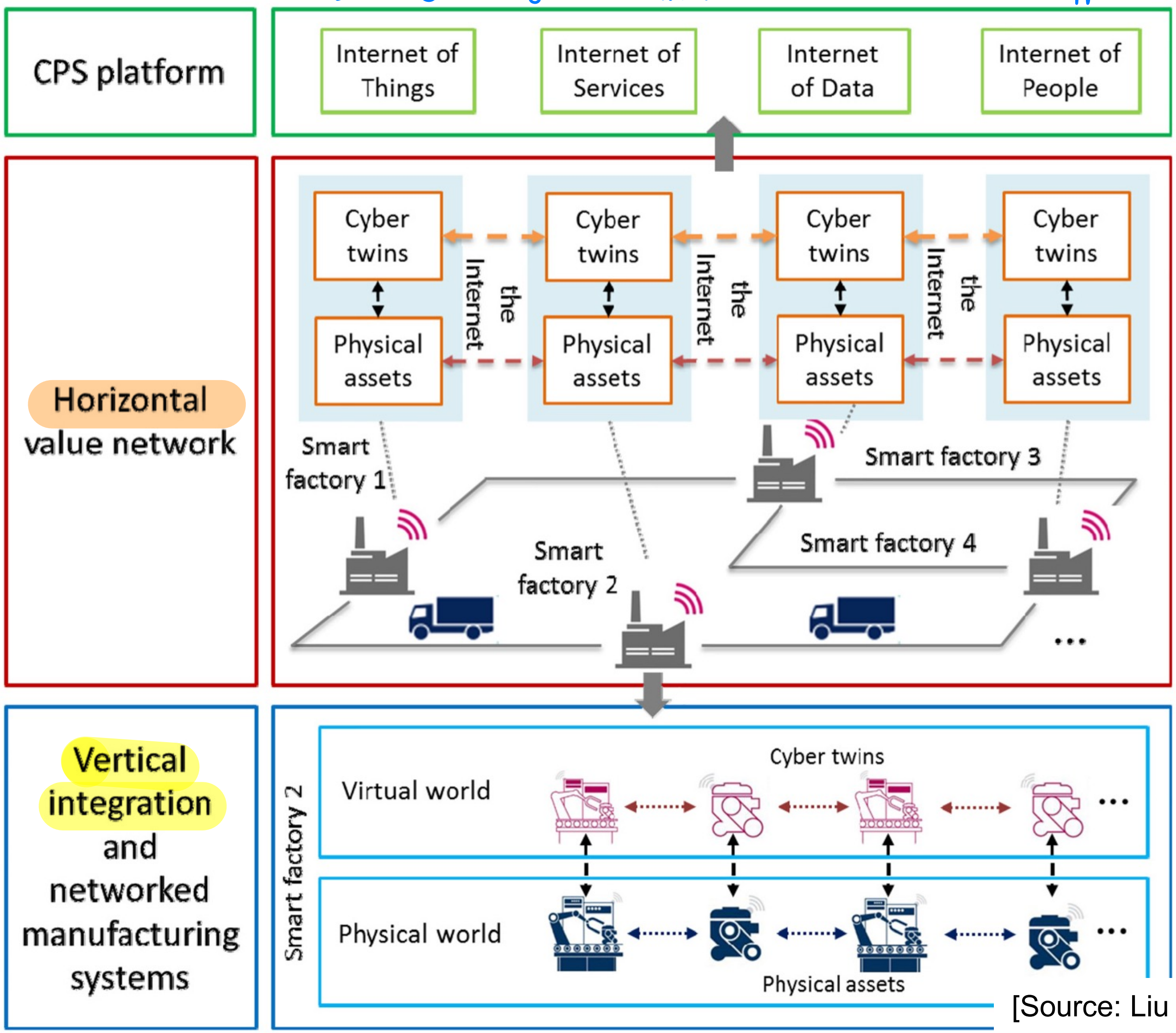
Horizontal and Vertical Integration

related to the plant (production environment)
↓
TRY TO LINK
digital representation to real
(DIGITAL TWIN) → SIMULATING Real plant

Simulation of behaviour to know what will happen → to synchronize!

many · env
↑
LINK
↓
Digital twin

without LINK...
you have a stand alone
represent. VIRTUAL
NOT USEFULL,
to perform simulation
but ISOLATED!



LINK INFORMATION
from sensors to
model
↓
for predicting
behaviour /
failure
(FUTURE)

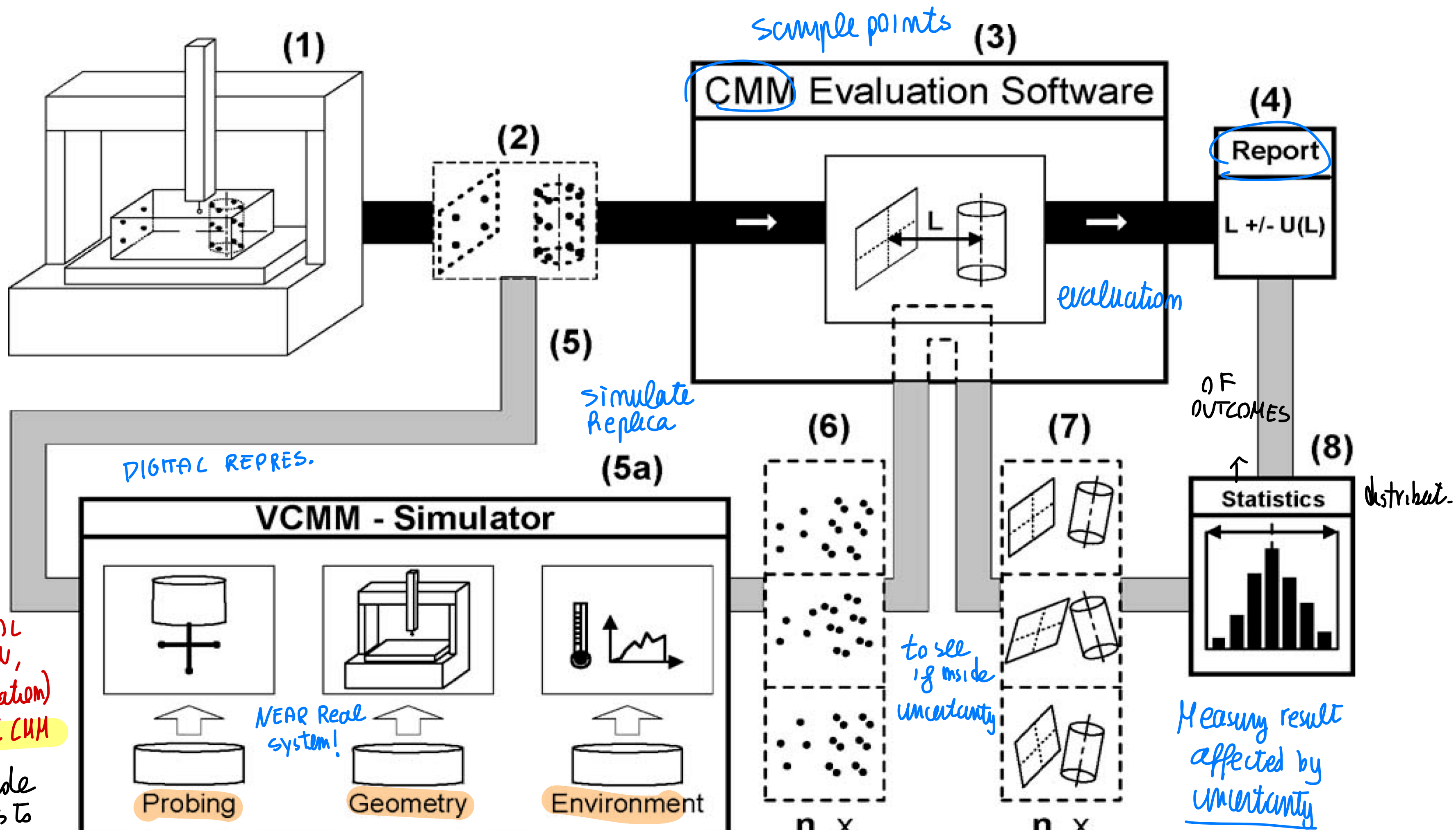
each company
should be
interconnected
in between
(HORIZONTAL
INTEGR.)

but we
want to
discuss about
different supplier
in the chain...
many actor in
the full process
↓
full INTEGRATION
Network, by
real ⇔ VIRTUAL
factory updated
↑
to take decisions
easily also for scheduling

[Source: Liu and Xu 2016]

SIMULATION / DIGITAL TWIN

(EXAMPLE)
to take decision about PART QUALITY...



DIGITAL TWIN, (simulation) VIRTUAL CMM
you decide points to sample... machine with possible ERRORS due to environments, ecc.. SIMULATE all errors combination

model of ERRORS

(generate many POINTS applying RANDOMLY errors!)

VCMM = Virtual Coordinate Measuring Machine

∀ cards random evaluate situation

ADVANCED MANUFACTURING SOLUTIONS

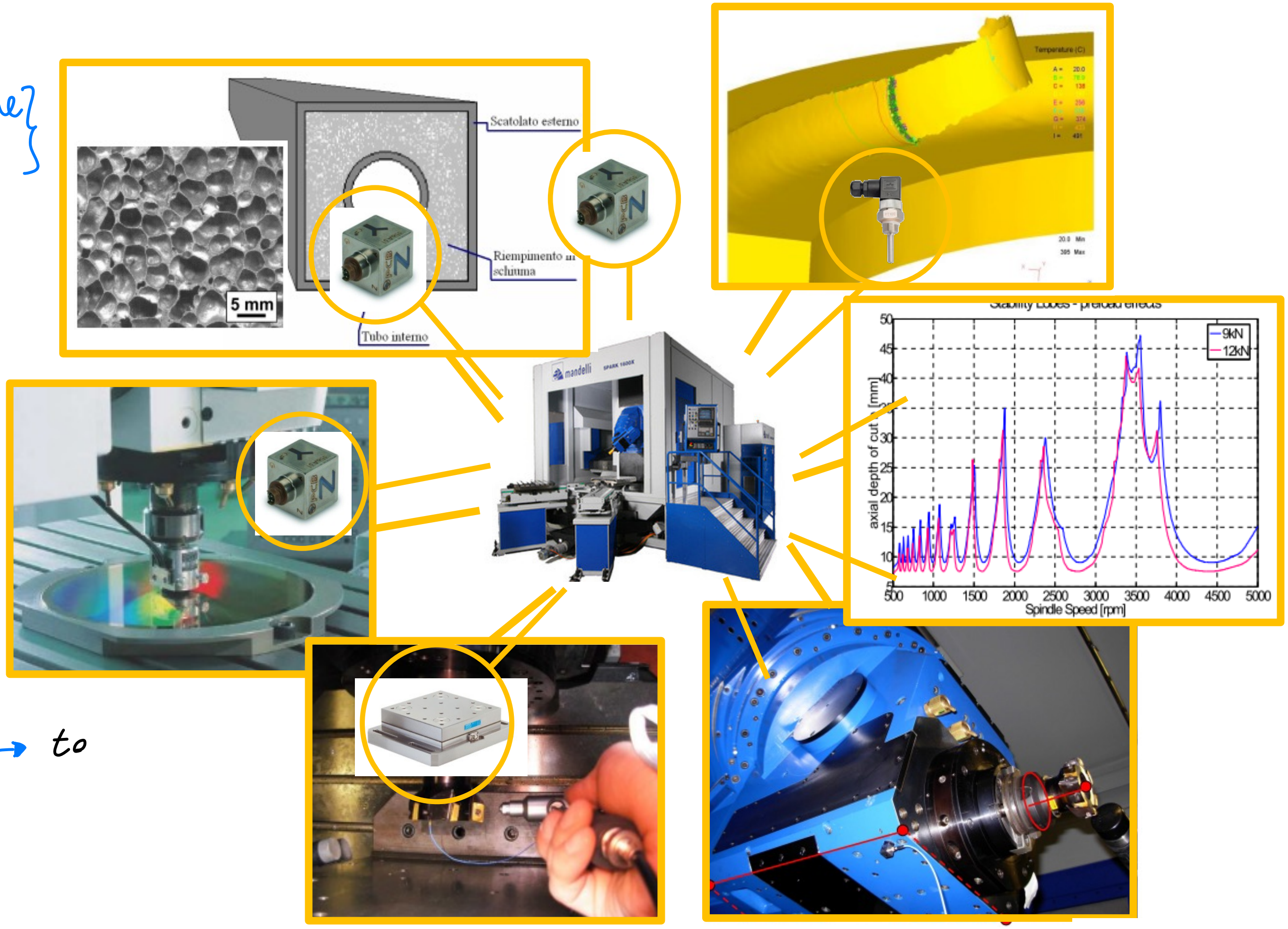
MACHINING

(collaborative Robots)

Machine Tools and Sensors: {Sensorize machine to have INFO}

- Power
- Force
- Vibrations
- Acoustic emission
- Images
- Videos (IR/visible range)

use info from sensors → to improve the output ⇒

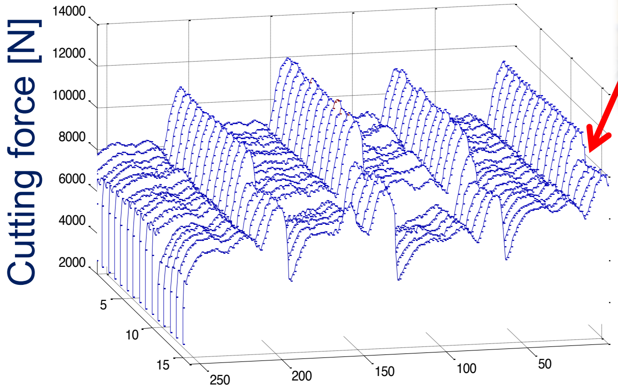


[Source: Polimi]

Advanced Manufacturing Solutions

MONITOR cutting FORCE

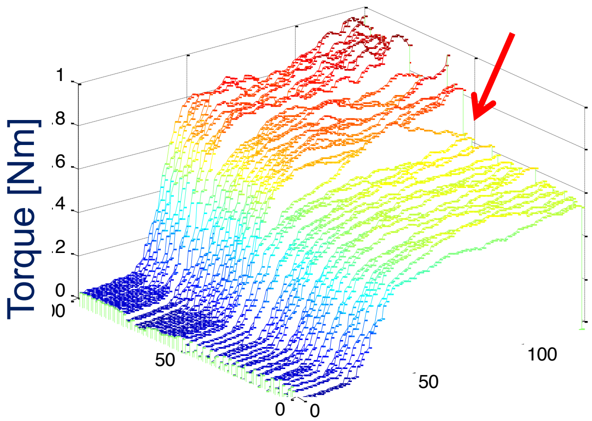
Milling hard-to-cut materials



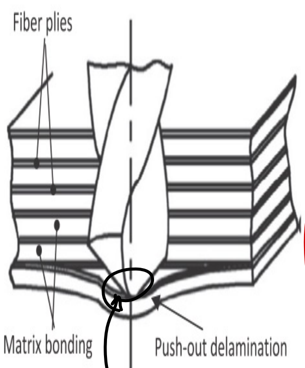
see cutting profile...
JUST look to F_c to see variation respect desired feature (MONITORING)

Tapping monitoring

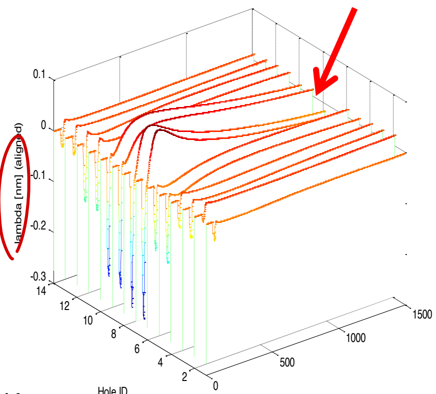
torque variation issue



Drilling of multi-layer materials (aerospace)

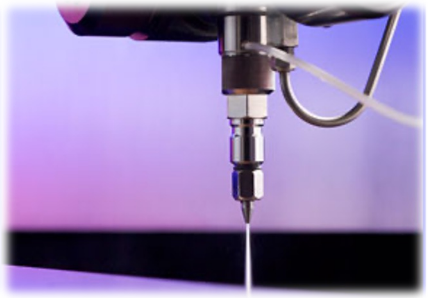


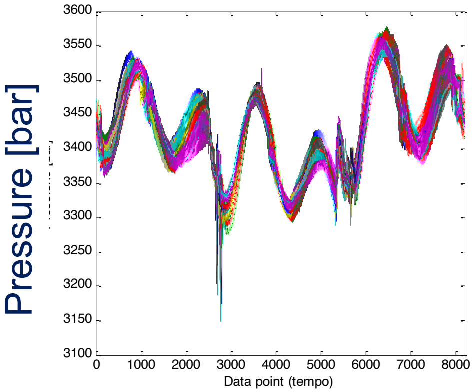
look out put to identify issue



MOOVING PORTION, lammination issue, NOT CUTTING → DEFECTS!

Waterjet and Abrasive Waterjet cutting





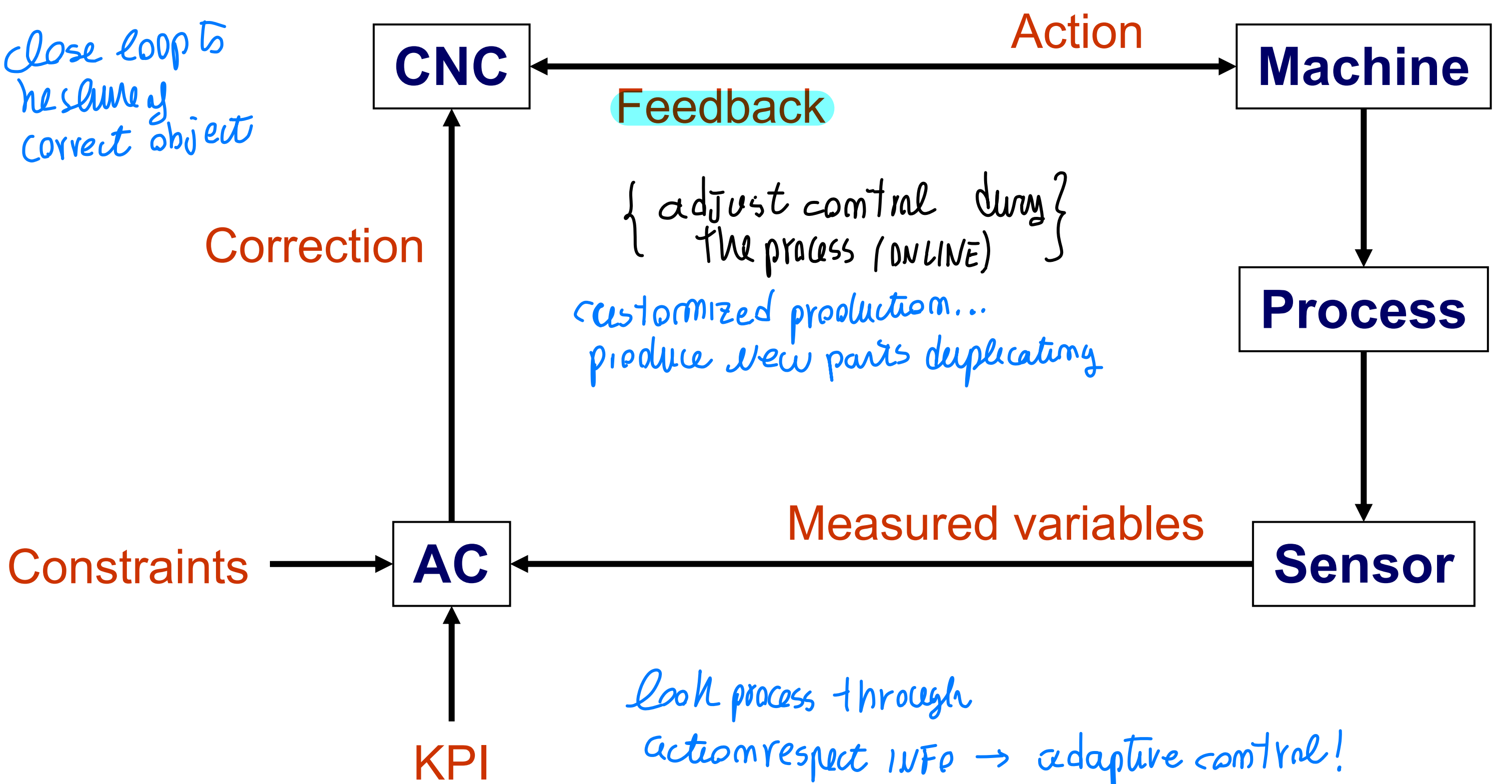
[Source: Polimi]

ADAPTIVE CONTROL - FRAMEWORK

Control scheme that adjusts the control system parameters from conditions detected during the process (ISO 2806:1994).

using SENSORS to control, look the process and understand what happens → close a loop! smart

close loop to be sure of correct object



{adjust control during the process (ONLINE)}

customized production... produce new parts duplicating

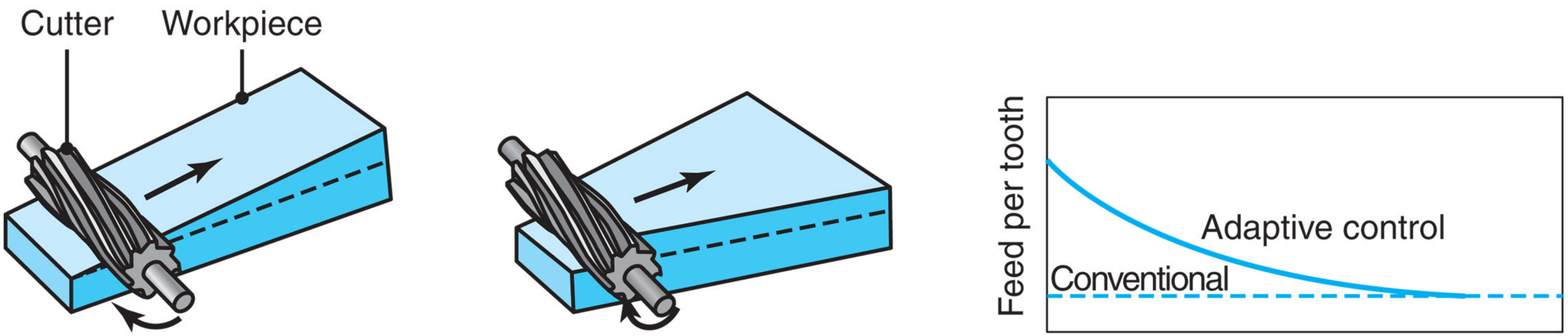
look process through action respect info → adaptive control!

use sensors to do that (low cost sensors good enough)

ADAPTIVE CONTROL - EXAMPLE

Adaptive control needs in slab milling.

⇓ what could happen
(material variability)



(a) Variable depth of cut (b) Variable width of cut (c) Cutter travel

variable depth of cut... ecc
ISSUE

IF you know issue you consider WORST CASE POWER Requirements!
from WORST CASE → conservative definition
ADAPT cutting TOOL → close the loop ⇒ change feed for faster cutting

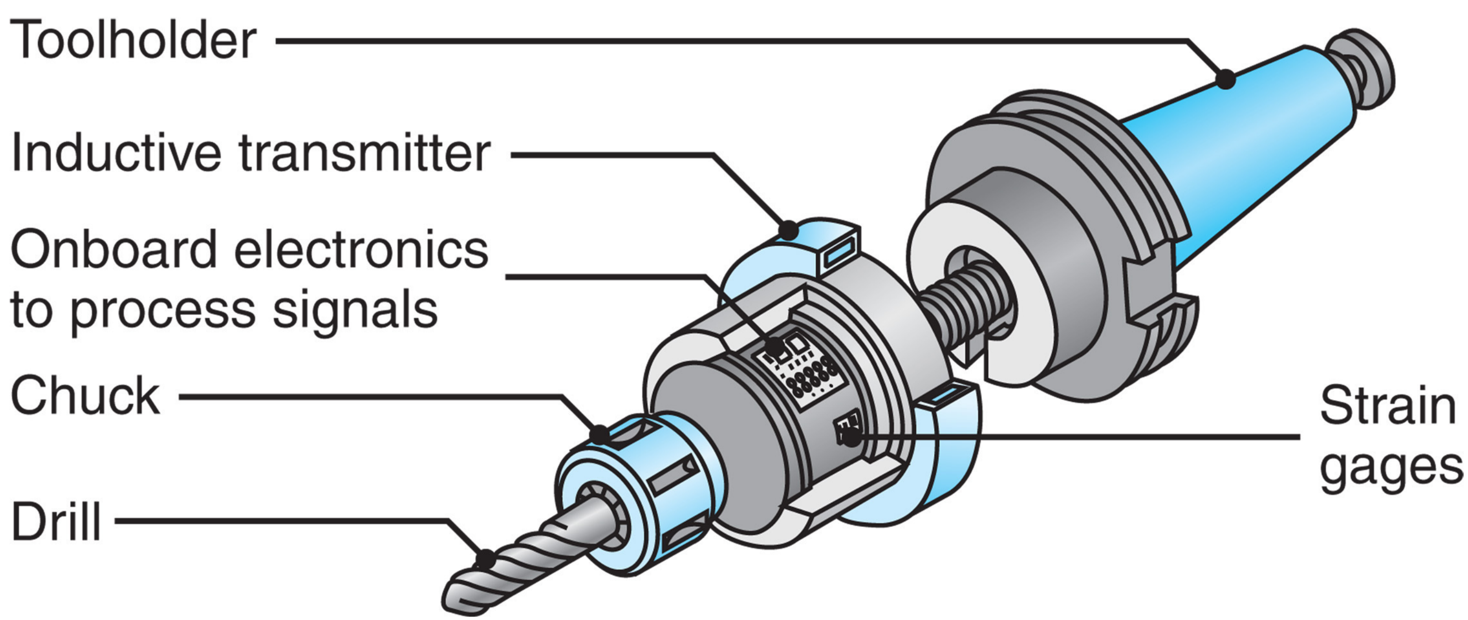
Adaptive Control and Sensors

Sensor

Unit which is actuated by a physical quantity and which gives a signal representing the value of that physical quantity (ISO 2806:1994).

collecting INFO...

mounted on the TOOL HOLDER



A toolholder equipped with thrust-force and torque sensors (*smart toolholder*).

collect information and transmit it!

WIRELESS communication

carefull to influence of magnetic fields!

ADAPTIVE CONTROL - APPROACHES

Control scheme that adjusts the control system parameters from conditions detected during the process (ISO 2806:1994).

3 main kind

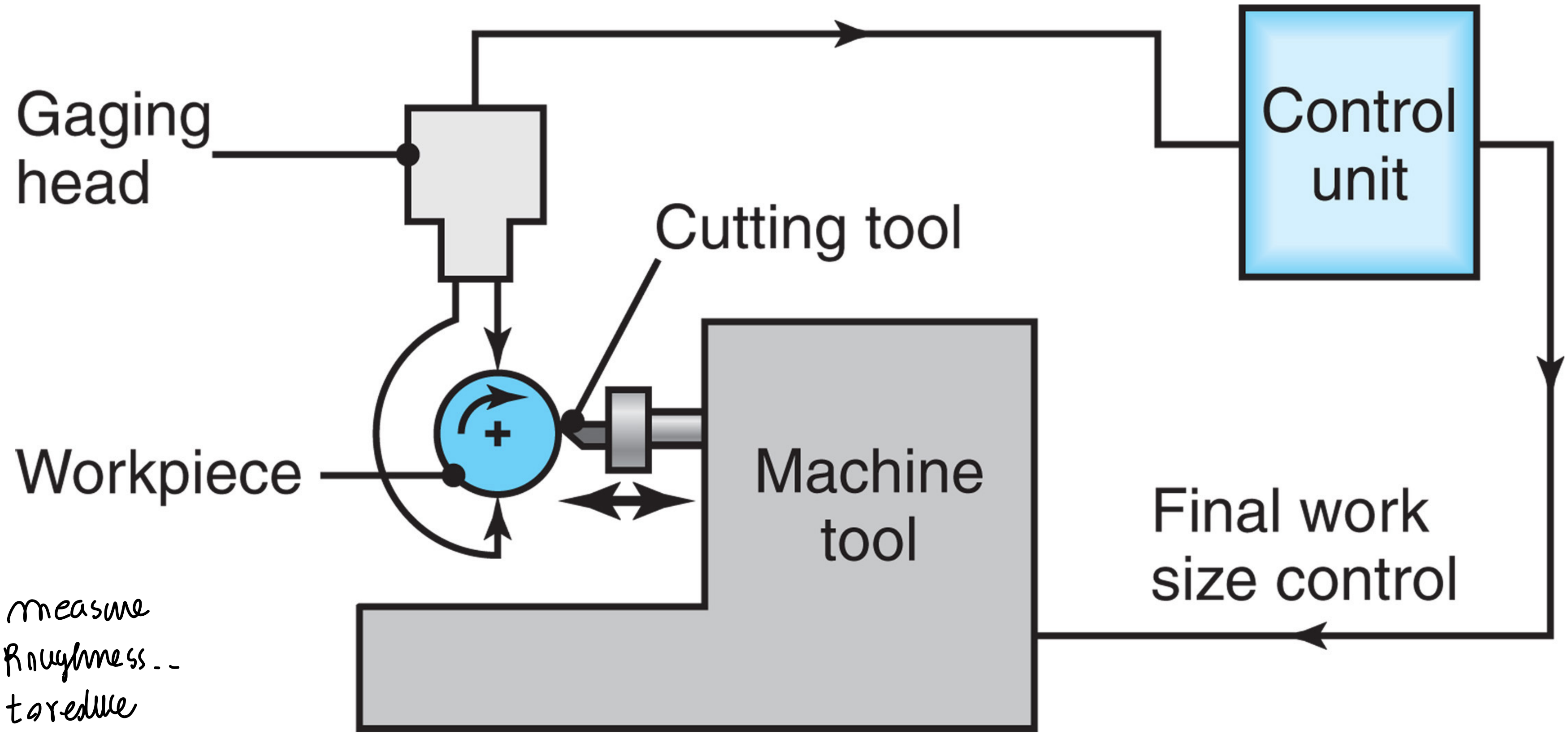
- GAC - geometric adaptive control
- ACC - adaptive control constraint
- ACO - adaptive control optimization

GEOMETRIC ADAPTIVE CONTROL

look to the PART during the making -- mapping Geometrical defects through measuring system

GAC aims at keeping tolerances and surface finishing in given ranges.

measure Diam during ROTATIONAL CUTTING → change something IF there is ISSUE during CUTTING



measure Roughness -- to reduce vibration

In-process inspection of workpiece diameter in a turning operation.

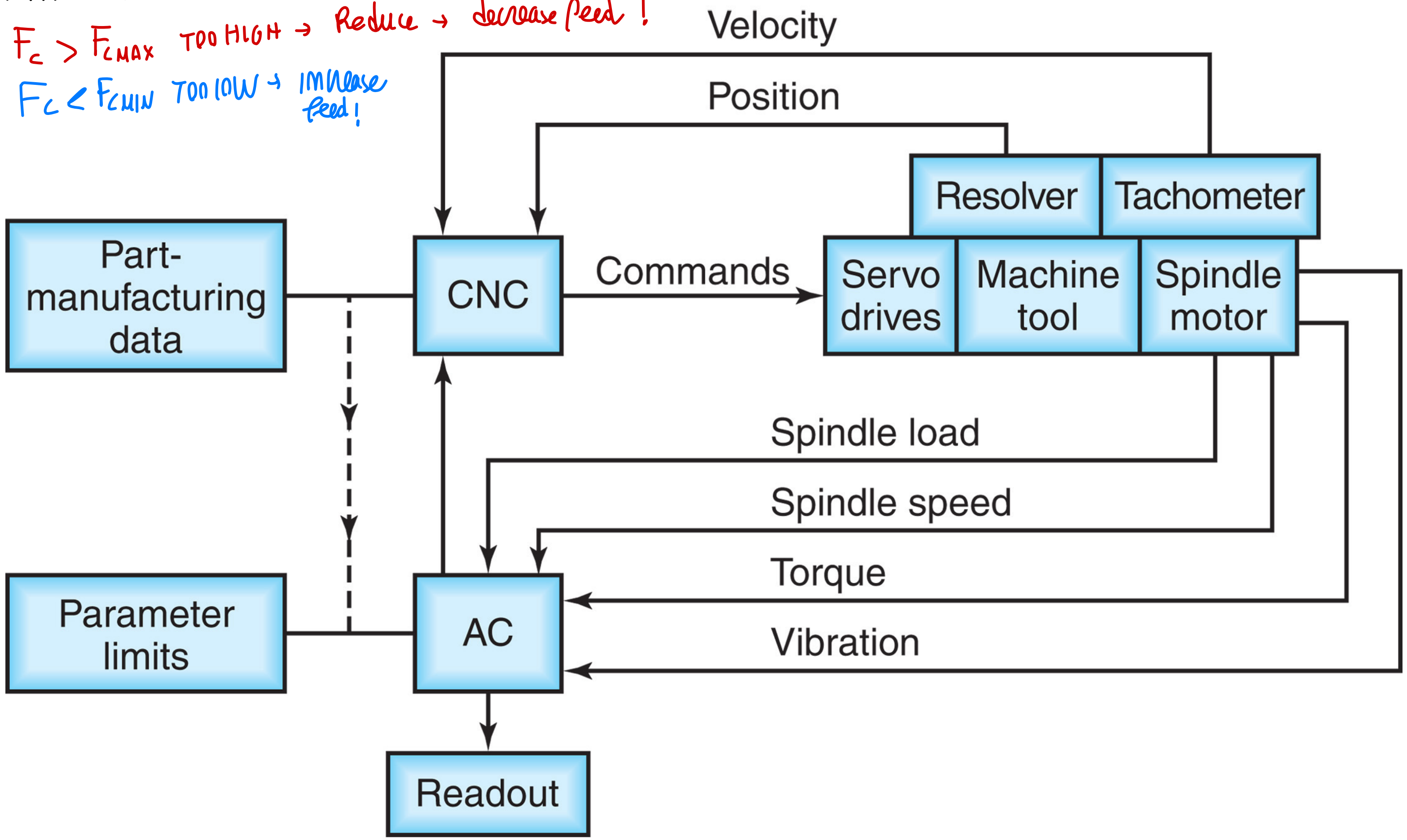
ADAPTIVE CONTROL CONSTRAINT

more sensors to GET INFO, to remain inside a given INTERVAL...

IF measure F_c that we want in an INTERVAL, we can adjust the value if OUTSIDE Range!

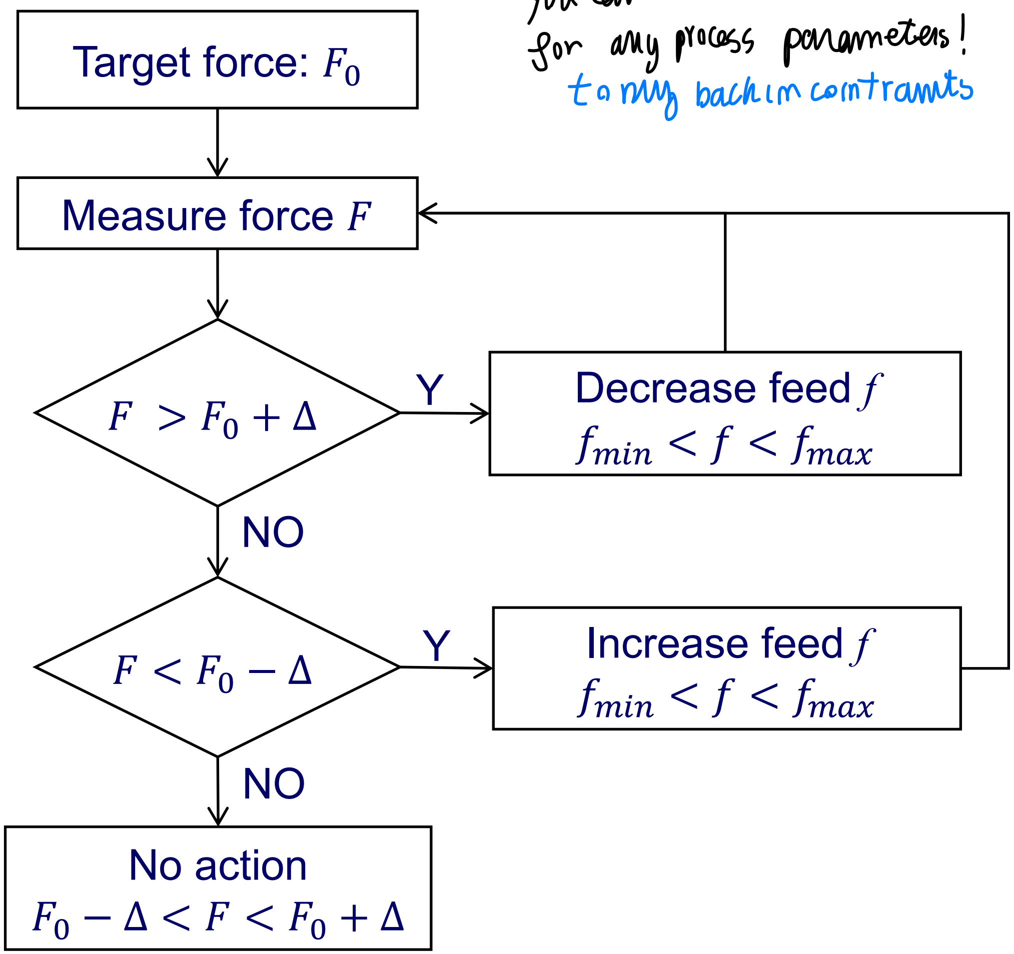
$F_c > F_{c\,MAX}$ TOO HIGH → Reduce → decrease feed!

$F_c < F_{c\,MIN}$ TOO LOW → increase feed!



ADAPTIVE CONTROL CONSTRAINT

you can do this
for any process parameters!
to any bachin contraints



ACC aims at keeping a technological variable in a given range satisfying the constraints of the technological system.

Typical variables are:

- Temperature
- Cutting Force
- Cutting Power
- Width of vibrations

ADAPTIVE CONTROL OPTIMIZATION

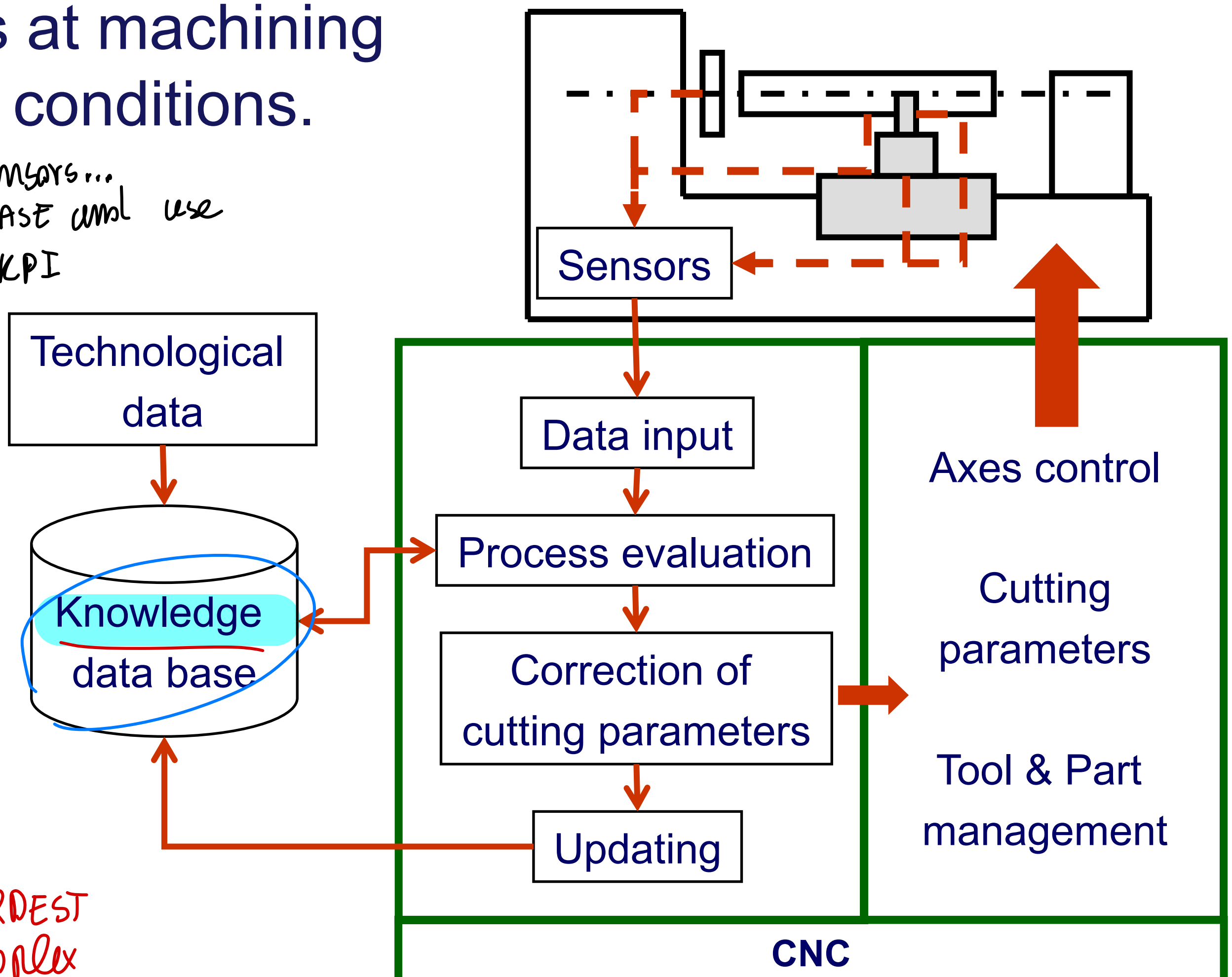
(Complex control)

ACO aims at machining at optimal conditions.

Collect INFO from sensors... KNOWLEDGE DATA BASE and use optimiz. according to KPI

acting on the machine

Generic description
↓
change process parameters
Re-planning
what to do!



HARDEST
complex
approach

MONITOR the state, complex to close the loop!

WORK IN a safe Range!

Lot's of innovation coming from '90 age...
AUTOMATION INNOVATION...